University Scholarship Allocation System

1. Problem Statement

The University Scholarship Allocation System is designed to evaluate students based on their academic performance, subject-wise marks, attendance percentage, and family annual income. It identifies eligible students, ranks them according to predefined rules, and allocates scholarships fairly.

2. Objective

- Evaluate students using aggregate marks.
- Check minimum marks in each subject.
- Verify attendance percentage.
- Check family annual income.
- Identify eligible students.
- Rank eligible students based on merit.
- Resolve ties using predefined rules.

3. Eligibility Criteria

Criteria	**Requirement**
Aggregate marks	75% or above
Minimum subject marks	60% in every subject
Attendance	75% or above
Annual family income	₹2,50,000 or below

4. Computational Thinking

1. Decomposition: Divide the problem into input collection, eligibility checking, ranking, and scholarship allocation.
2. Pattern Recognition: Identify common eligibility conditions among students.
3. Abstraction: Consider only marks, attendance, and income.
4. Algorithm Design: Develop steps to evaluate and rank students.

5. Algorithm

1. Start.

2. Read the number of students.
3. Read each student's name, subject marks, attendance, and annual income.
4. Calculate aggregate percentage.
5. Check minimum subject marks.
6. Check attendance and income criteria.
7. If all conditions are satisfied, mark the student as eligible.
8. Store eligible students in a list.
9. Sort eligible students using:
    - Higher aggregate marks first.
    - Higher attendance next.
    - Lower family income next.
    - Student name alphabetically if all other criteria are equal.
10. Display the merit list.
11. Allocate scholarships according to rank.
12. Stop.

6. Pseudocode

```
START

INPUT number_of_students

CREATE empty eligible_list

FOR each student

    INPUT student_name

    INPUT subject_marks

    INPUT attendance

    INPUT annual_income
```

```
    CALCULATE aggregate_marks

    IF all subject marks >= 60 AND
      aggregate_marks >= 75 AND
      attendance >= 75 AND
      annual_income <= 250000 THEN

        ADD student details to eligible_list

    ELSE
        DISPLAY "Not Eligible"
    END IF

END FOR

SORT eligible_list BY
    aggregate_marks DESCENDING,
    attendance DESCENDING,
    annual_income ASCENDING,
    student_name ASCENDING

DISPLAY merit_list

FOR each student in eligible_list

    IF rank == 1
       DISPLAY "First Scholarship"
```

```
    ELSE IF rank == 2
        DISPLAY "Second Scholarship"
    ELSE IF rank == 3
        DISPLAY "Third Scholarship"
    ELSE
        DISPLAY "Waiting List"
    END IF

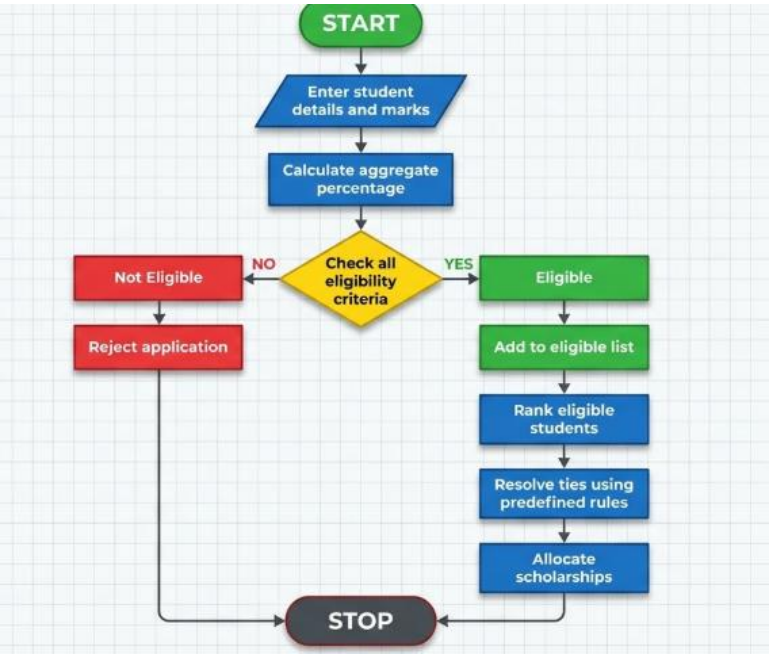
END FOR

STOP
```

7. Python Program

```
students = []eligible = []n = int(input("Enter number of students: "))for i in range(n): print("\nStudent", i + 1)  name = input("Enter student name: ")  marks = []  for j in range(3):  m = int(input("Enter subject marks: "))  marks.append(m)  attendance = float(input("Enter attendance percentage: "))  income = int(input("Enter annual family income: "))  total = sum(marks)  aggregate = total / len(marks)  if min(marks) >= 60 and aggregate >= 75 and attendance >= 75 and income <= 250000:  student = (name, aggregate, attendance, income)  eligible.append(student)  else:  print(name, "is Not Eligible")# Ranking and tie-breakingeligible.sort(key=lambda x: (-x[1], -x[2], x[3], x[0]))print("\n--- SCHOLARSHIP MERIT LIST ---")for i, student in enumerate(eligible):  name, marks, attendance, income = student  if i == 0:  award = "First Scholarship"  elif i == 1:  award = "Second Scholarship"  elif i == 2:  award = "Third Scholarship"  else:  award = "Waiting List"  print("\nRank:", i + 1)  print("Name:", name)  print("Aggregate:", marks)  print("Attendance:", attendance)  print("Income:", income)  print("Award:", award)
```

8. FlOWCHART:



START

Enter student details and marks

Calculate aggregate percentage

Check all eligibility criteria

YES

Eligible

Add to eligible list

NO

Not Eligible

Reject application

Rank eligible students

Resolve ties using predefined rules

Allocate scholarships

STOP

9. Concepts Used

Python Concept	**Application**

Variables	Store student details
Data types	Integer, float, string
Operators	Calculate aggregate
Conditional statements	Check eligibility
Loops	Process multiple students
Strings	Store student names
Lists	Store subject marks
Nested lists	Store multiple student records
Tuples	Store eligible student details
Tuple unpacking	Retrieve student information
Pattern generation	Display merit list using numbered ranks

10. Expected Output

--- SCHOLARSHIP MERIT LIST ---

Rank: 1

Name: Ravi

Aggregate: 90.0

Attendance: 95.0

Income: 150000

Award: First Scholarship

Rank: 2

Name: Priya

Aggregate: 85.0

Attendance: 90.0

Income: 180000

Award: Second Scholarship

Rank: 3

Name: Kumar

Aggregate: 80.0

Attendance: 85.0

Income: 200000

Award: Third Scholarship

11. Conclusion

The University Scholarship Allocation System evaluates students based on academic marks, attendance, and family income. It identifies eligible candidates, prepares a merit list, resolves ties, and allocates scholarships fairly using Python programming concepts.